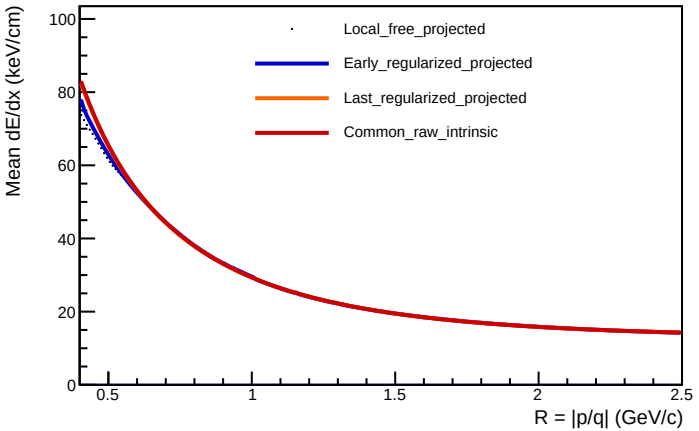
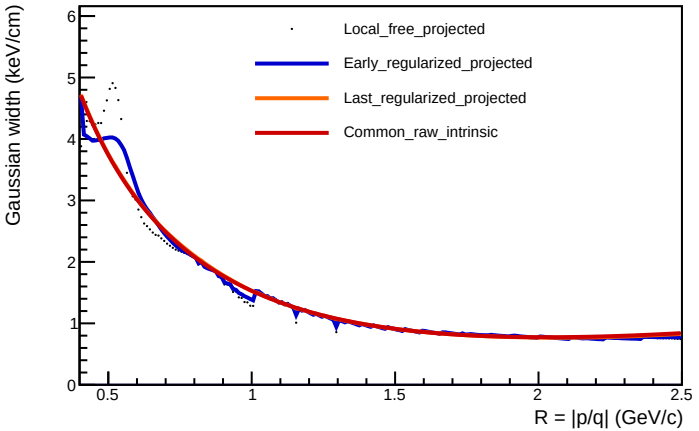
## $^3$He component mean

Study only: curve smoothness is NOT isotope validation



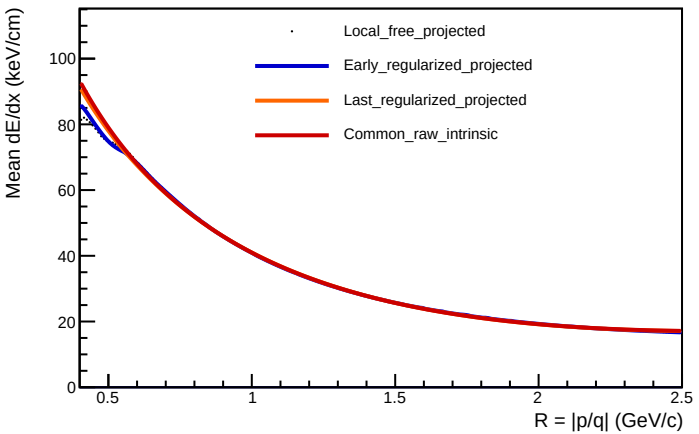

## $^3$He component width

Study only: curve smoothness is NOT isotope validation



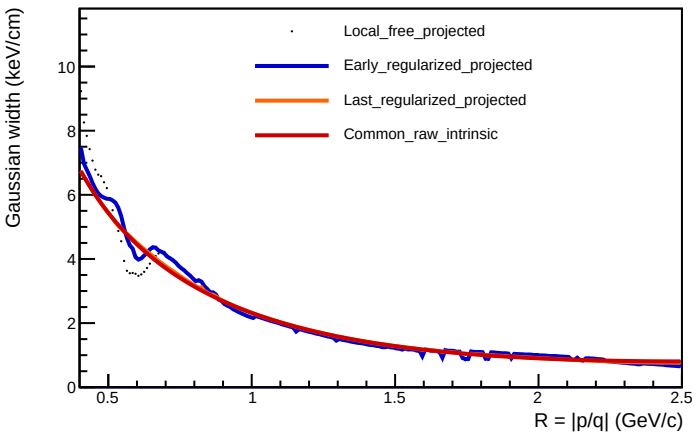

## $^4$He component mean

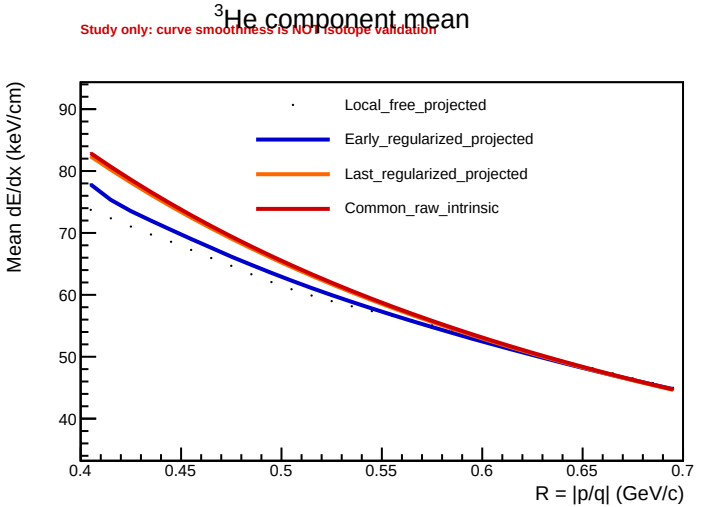
Study only: curve smoothness is NOT isotope validation

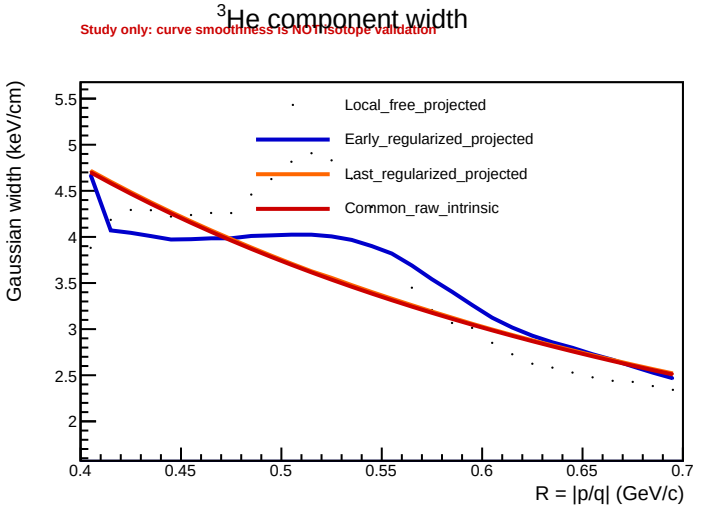


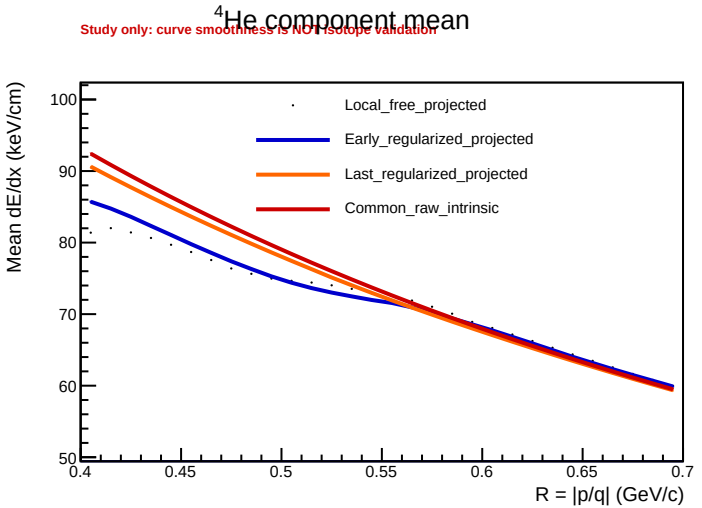

## $^4$He component width

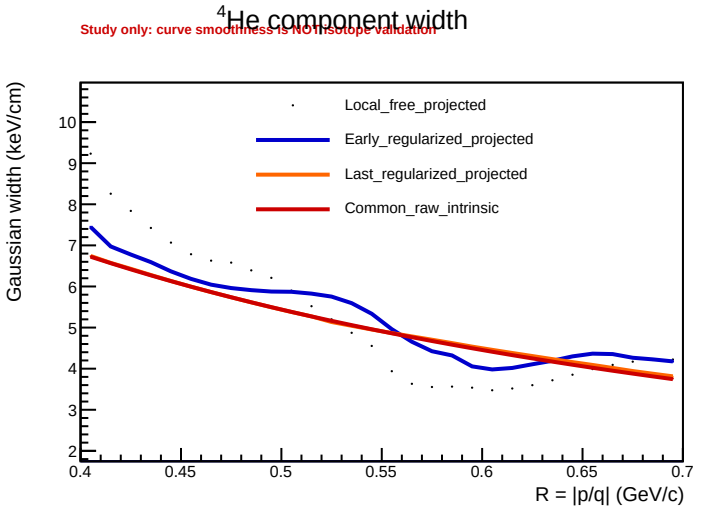
Study only: curve smoothness is NOT isotope validation




$^3$He component mean
Study only: curve smoothness is NOT isotope validation
Mean dE/dx (keV/cm)
R = |p/q| (GeV/c)
Local_free_projected
Early_regularized_projected
Last_regularized_projected
Common_raw_intrinsic

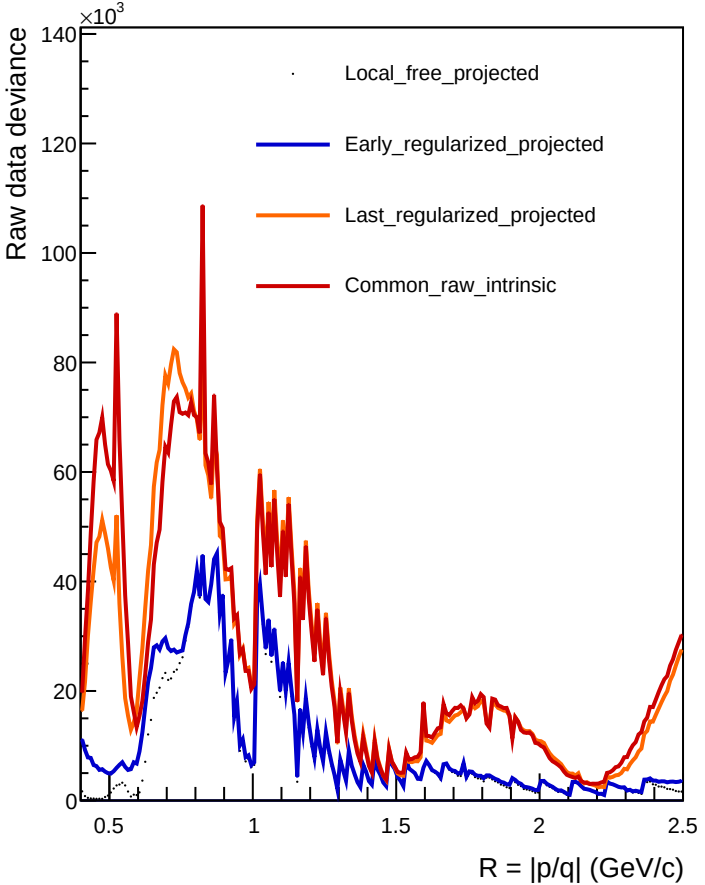

$^3$He component width
Study only: curve smoothness is NOT isotope validation
Gaussian width (keV/cm)
R = |p/q| (GeV/c)
Local_free_projected
Early_regularized_projected
Last_regularized_projected
Common_raw_intrinsic


$^4$He component mean
Study only: curve smoothness is NOT isotope validation
Mean dE/dx (keV/cm)
R = |p/q| (GeV/c)
Local_free_projected
Early_regularized_projected
Last_regularized_projected
Common_raw_intrinsic


$^4$He component width
Study only: curve smoothness is NOT isotope validation
Gaussian width (keV/cm)
R = |p/q| (GeV/c)
Local_free_projected
Early_regularized_projected
Last_regularized_projected
Common_raw_intrinsic

## Unpenalized data fit comparison

**Study only: curve smoothness is NOT isotope validation**




## Conditional residual diagnostic (not global fit NDF)

**Study only: curve smoothness is NOT isotope validation**

